

Selecting efficient pharmaceutical binder for developing robust extended-release matrices of Compritol® 888 ATO prepared using wet granulation technique: Part 1 Studying impact on granule & tablet properties

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OBJECTIVE:

This study evaluates the binding efficiency of four commonly used pharmaceutical binders to hold hydrophilic API's of different solubilities with hydrophobic, waxy ER matrices of Compritol® 888 ATO as sustained release agent by studying their impact on the granule and tablet properties formulated using the same.

METHODS:

Physicochemical Properties of binders under study

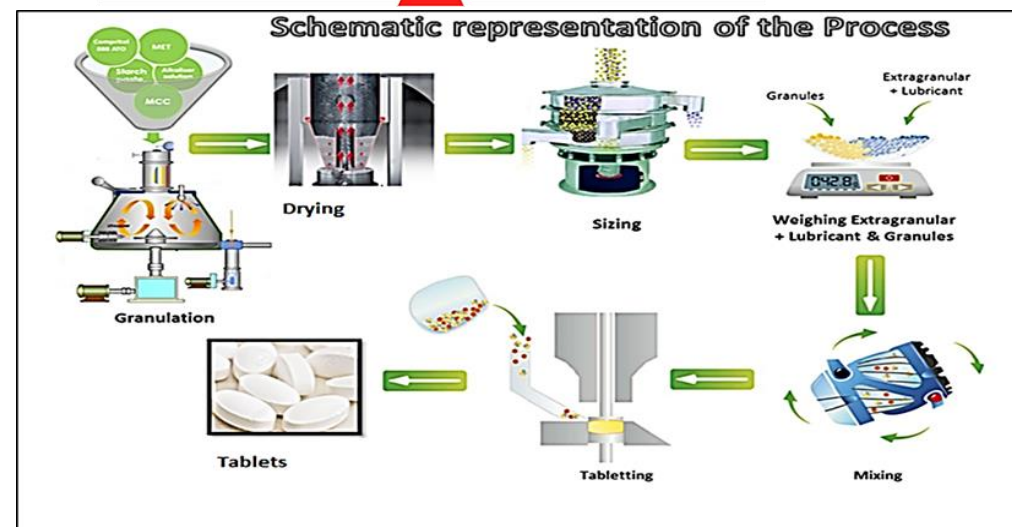
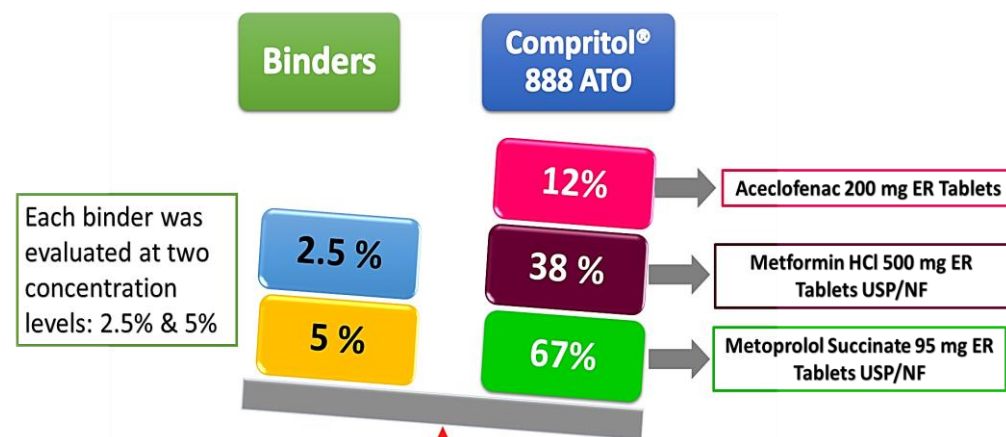
Excipients Properties	HPC L	HPMC E6	PVP K30	PVP K90
Chemical name	Hydroxy propyl cellulose	Hydroxy propyl methyl cellulose	poly(1-vinyl-2-pyrrolidone)	poly(1-vinyl-2-pyrrolidone)
Avg. molecular weight	95,000 Daltons	21,000 Daltons	54,000 Daltons	15,00,000 Daltons
Viscosity (5% solution at 25°C,	105 cps	15 cps	3 cps	150 cps
Melting point(°C)/Tg	Soften at 130°C	170-180°C	130°C	130°C
Appearance in Water at Room Temperature	Viscous Gel	Viscous Gel	Clear Solution	Clear Solution

Physicochemical Properties of APIs & Formulation Characteristics

API Names	Aceclofenac	Metformin HCl	Metoprolol Succinate
Dose	200 mg	500 mg	95 mg
Log P	3.88	-0.92	1.76
Plasma half life	4 hours.	6.2 hours	3 to 7 hours.
Water solubility	Insoluble (pH dependent solubility)	> 50 mg/ml 200 mg/mL	Freely soluble
Concentration of Compritol® 888 ATO	12 %	38 %	67 %
Tablet weight	400 mg	1035 mg	850 mg
Tooling details	9.5 mm Round Standard Concave punch	19 X 9 mm Caplet punch	16 X 9 mm Caplet punch

Study Plan

ER tablet formulations for each API were prepared using following variations



RESULTS:

Aceclofenac 200 mg ER Tablets ; Compritol® 888 ATO: 12 %

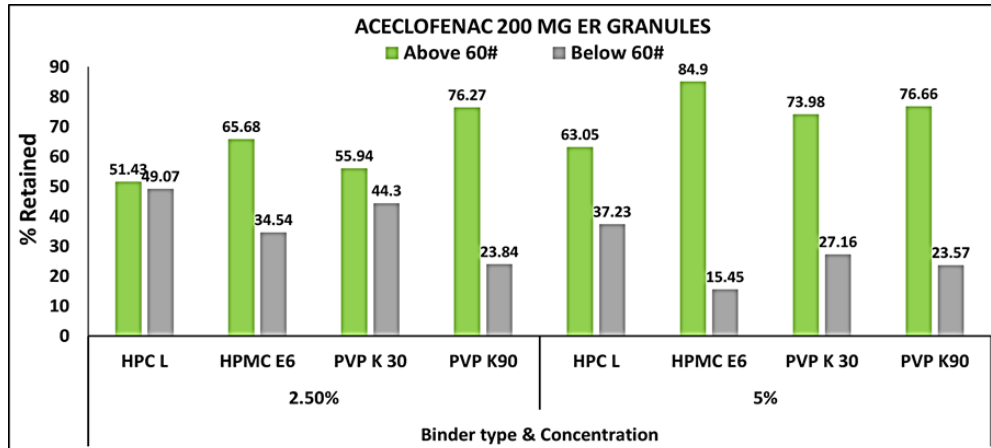


Fig. 1 Particle size distribution of Aceclofenac 200 mg ER granules prepared using binders under study

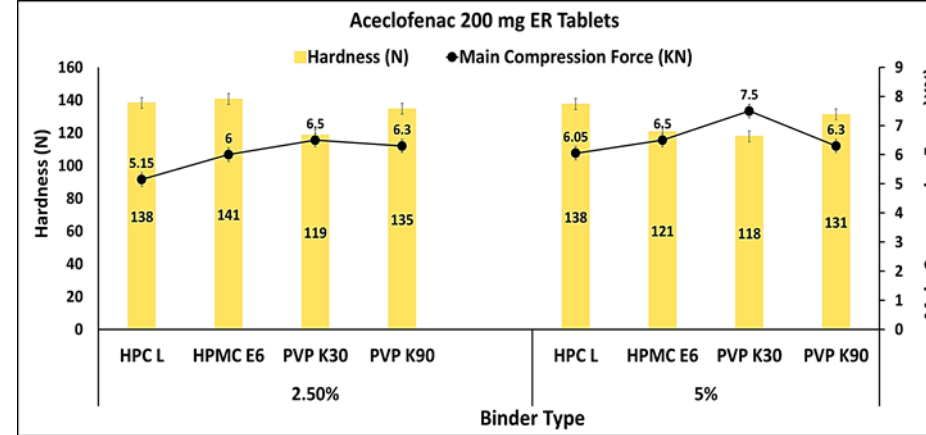


Fig. 2 Comparison of max. Hardness versus Main Compression force for Aceclofenac ER Tablets

- HPMC E6 at 5% & PVP K90 at 2.5% & 5% concentration showed better control over PSD of the granules.
- Higher tablet hardness was achieved with HPC L, HPMC E6 & PVP K 90 at 2.5% concentration.
- Main Compression Force was found to be same for all the batches.

Metformin HCl 500mg ER Tablets USP/ NF ; Compritol® 888 ATO: 38 %

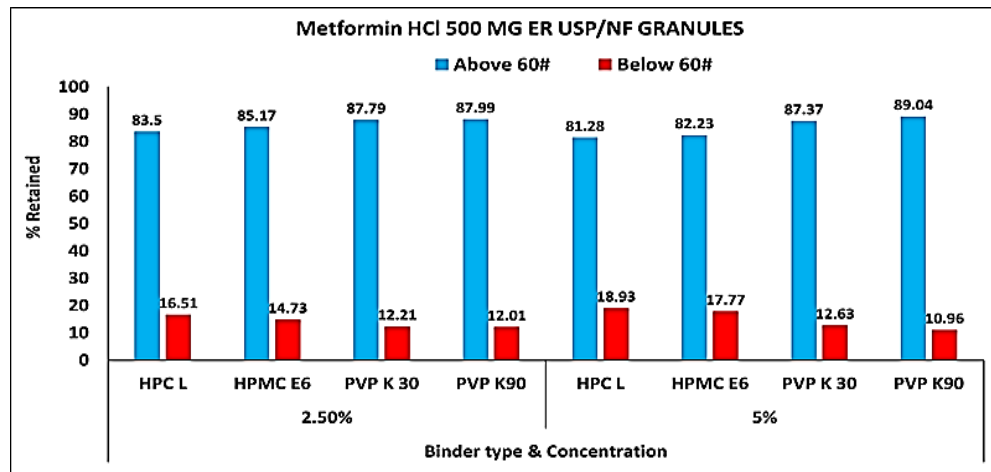


Fig. 3 Particle size distribution of Metformin HCl 500 mg ER granules prepared using binders under study

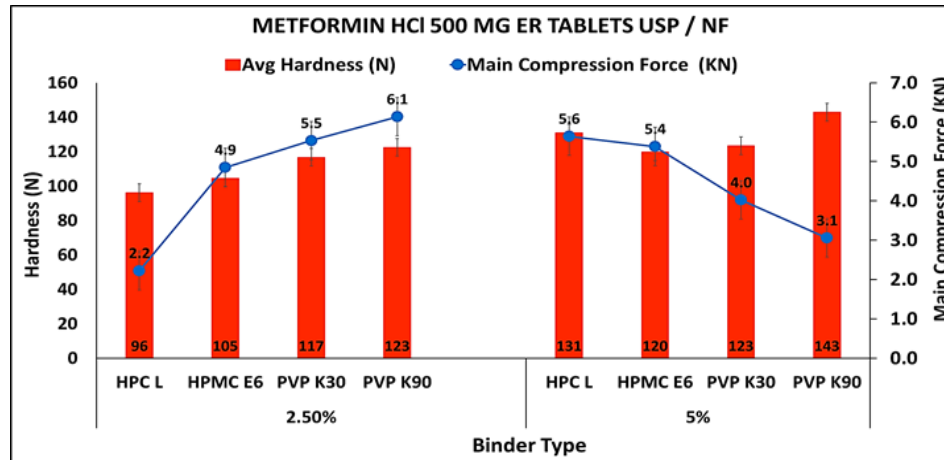


Fig. 4 Comparison of max. Hardness versus Main Compression force for Metformin HCl ER Tablets

- PVP K 30 & PVP K 90 at 2.5% & 5% concentration showed better control over PSD of the granules.
- Higher tablet hardness was achieved at lower Main compression force for formulations with PVP K90 at 5% concentration

Metoprolol Succinate 95 mg ER Tablets USP ; Compritol® 888 ATO: 67 %

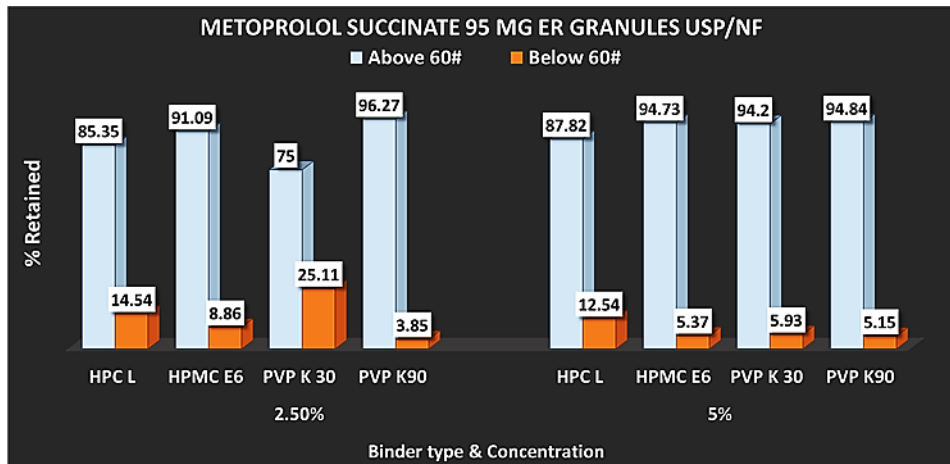


Fig. 5 Particle size distribution of Metoprolol Succinate 95 mg ER granules prepared using binders under study

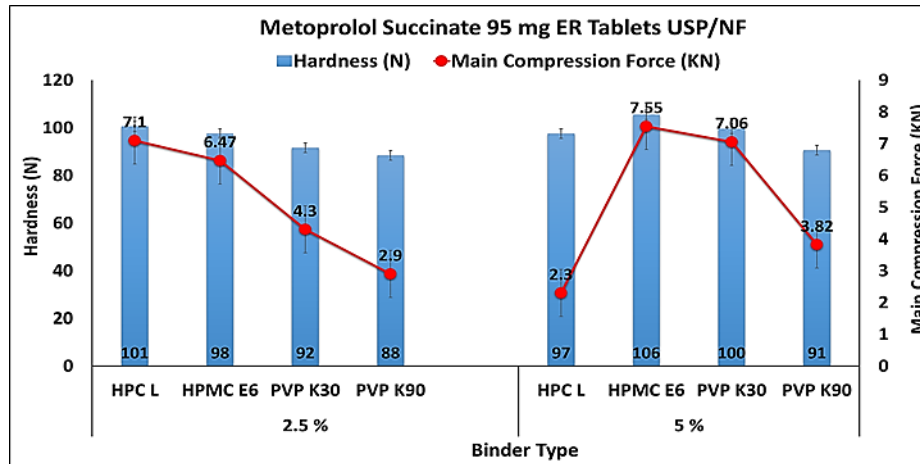


Fig. 6 Comparison of max. Hardness versus Main Compression force for Metoprolol Succinate ER Tablets

- PVP K 90 at 2.5% & 5% concentration while HPMC E6 & PVP K30 at 5% showed good control over PSD of the granules.
- Higher tablet hardness at low main compression force was achieved with PVP K90 & HPC L at 5 % concentration.

CONCLUSIONS:

1. From the study it was observed that binders with high viscosity gave best results for enhancing the binding between hydrophobic, waxy Compritol® 888 ATO and hydrophilic APIs.
2. Compritol® 888 ATO SR granules of all the three APIs prepared using PVP K90 showed good flow properties and enhanced control over particle size distribution even at lower concentrations.
3. Tablets of all the three APIs prepared using PVP K90 were able to achieve higher hardness values and showed good compaction behavior, at lower main compression forces.

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REFERENCES:

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2. Morkhade DM. Comparative impact of different binder addition methods, binders and diluents on resulting granule and tablet attributes via high shear wet granulation. Powder technology. 2017 Oct 1; 320:114-120.